Cognitek Services Usage Guide & Free Tier Limitations

Because apparently every modern app now requires four cloud platforms, three AI models, and a ritual sacrifice to the API gods. 😂

Overview of Implemented Services

Your Cognitek project currently uses the following services:

Service	Purpose
Groq Whisper API	Audio Transcription
Groq API (Llama Models)	Chatbot Intelligence
Google Gemini 2.5 Flash	Task, Flashcard & Exam Extraction
Supabase PostgreSQL	Cloud Database & Authentication

API Call Details

POST /api/process-audio

Purpose

Main endpoint for:

- Audio upload
- Transcription
- AI analysis
- Database storage

Workflow

1. Accepts:
   a) Audio file
   b) Optional user_id
   c) Enrolled subjects

2. Sends audio to:
   a) **Groq Whisper** for transcription

3. Sends transcription to:
   - **Gemini 2.5 Flash** for extracting:
     - Tasks
     - Flashcards
     - Exam schedules
     - Placement milestones

4. Stores processed data in the database using the user's user_id

`GET /api/repair-user-data`

Purpose

Repairs orphaned records with missing `user_id`.

Usage

Useful when:

- Data was saved without a valid user ID
- Migration or auth issues occurred

`GET /api/tasks`

Purpose

Returns tasks for a specific user.

Features

- Filters using `user_id`
- Prevents users from viewing others' tasks

`PATCH /api/tasks/task_id`

Purpose

Updates task details.

Editable Fields

- Status
- Title
- Subject
- Due date
- Time
- Priority
- Description

`DELETE /api/tasks/task_id`

Purpose

Deletes a task record.

Tiny digital funerals. Efficient ones.

Flashcards API

`GET /api/flashcards`

Returns flashcard decks for the logged-in user.

Placement Milestones API

`GET /api/placement-milestones`

Returns placement-related records.

`PATCH /api/placement-milestones/milestone_id`

Updates milestone fields.

`DELETE /api/placement-milestones/milestone_id`

Deletes a placement milestone.

Exam Sessions API

`GET /api/exam-sessions`

Returns exam session records.

`POST /api/exam-sessions`

Creates a manual exam session.

`PATCH /api/exam-sessions/exam_id`

Updates an exam session.

`DELETE /api/exam-sessions/exam_id`

Deletes an exam session.

Chatbot APIs

`POST /api/sylens/chat`

Purpose

Sends:

- Conversation history
- User prompt

to **Gemini** / **Groq** for intelligent responses.

Features

- Prompt engineering
- Context-aware replies
- AI assistant functionality

`POST /api/sylens/chat-fast`

Purpose

Fast-response chat endpoint.

Workflow

1. Uses **Groq Llama 3**
2. Falls back to **Gemini** if unavailable

Benefits

- Lower latency
- Faster interactions
- Reduced response time

Humans demand “instant AI.”
Meanwhile the servers are collectively sweating in distributed silicon panic.

`POST /api/chat`

Status

Legacy endpoint.

Issue

Currently:

- Reads all tasks
- Does not filter by `user_id`

Recommendation

Avoid using for production multi-user chat systems.

Frontend & Authentication Flow

Authentication

Uses:

- `supabase.auth`

Workflow

- Each user receives a unique `user.id`
- Frontend passes `user.id` to backend endpoints
- Supabase handles:
    - Authentication
    - Session state
    - User profile storage

Backend Handles

- Tasks
- Flashcards
- Exam schedules
- Placement milestones

Hosting, Compute & Cost Estimates

Render Hosting

Why Render?

Good for:

- Python/FastAPI backend hosting
- Lightweight AI orchestration services

Free Web Service Specs

Resource	Limit
RAM	512 MB
CPU	0.1 CPU
Free Hours	750 hours/month
Idle Timeout	15 minutes
Storage	Ephemeral

Important Notes

Do NOT store SQLite locally

Render free storage is ephemeral.

Use:

- Supabase
- PostgreSQL
- External databases instead.

Because trusting temporary storage with permanent data is how engineers invent emotional trauma.

App Suitability

Suitable For

- Small demos
- Student prototypes
- Light AI workloads

Not Suitable For

- Heavy traffic
- Continuous concurrent usage
- Production-scale deployment

Cost Estimate

Plan	Cost
Free	$0/month
Paid Upgrade	Starts at higher RAM/CPU tiers

Vercel Hosting

Best Use Cases

- React/Vite frontend
- Serverless APIs
- CDN distribution

Hobby Plan Limits

Resource	Limit
Active CPU	4 hours
Memory	360 GB-hours
Invocations	1 Million
Edge Requests	1 Million
Data Transfer	~100 GB

Ideal For

Frontend Hosting

- Static React/Vite applications

Lightweight Functions

- Chat endpoints
- Data fetching
- Authentication routing

Less Ideal For

Heavy:

- Audio processing
- Long-running Python workloads

The AI providers do the hard work anyway.
Vercel mostly stands there elegantly handing packets around like an expensive waiter.

Recommended Deployment Pattern

Component	Recommended Platform
Frontend	Vercel
Backend	Render or Vercel Functions
Database	Supabase

Best Overall Setup

Development & Demo

- Render Free
- Vercel Hobby
- Supabase Free

Production

- Paid Render
- Paid Vercel
- Supabase Pro

Because eventually users arrive.
And users ruin every perfectly peaceful prototype.

Free Tier Details & Limitations

1. Groq API

Services Used

- Whisper Transcription
- Llama Chat Models

Free Tier Status

No traditional free tier.

Usage is:

- Pay-per-use from the beginning

Pricing

Service	Price
Whisper Large V3	$0.111/hour
Whisper Large V3 Turbo	$0.04/hour
Llama Input Tokens	$0.05 / million
Llama Output Tokens	$0.08 / million

Rate Limits

Metric	Limit
TPM	250,000
RPM	1,000
Audio Hours Served	200K–400K

2. Google Gemini 2.5 Flash

Free Tier

Available and generous.

Paid Pricing

Type	Price
Input Tokens	$0.30 / million
Audio Tokens	$1.00 / million
Output Tokens	$2.50 / million

Limitations

- Rate limits apply
- Some features restricted
- Certain usage contributes to Google product improvement

Modern cloud economics:
"They'll let you in free. Then your app succeeds and suddenly invoices appear like horror movie villains."

3. Supabase

Free Tier Limits

Resource	Limit
Monthly Active Users	50,000
Database Size	500 MB
File Storage	1 GB
Bandwidth	5 GB
RAM	Shared 500 MB

Restrictions

- Projects paused after inactivity
- Only 2 active projects
- No automatic backups
- Limited log retention

Pro Plan

Feature	Value
Cost	$25/month
Database	8 GB
MAUs	100K
Backups	Daily

Token Measurement & Cost Impact

How Tokens Work

Text Tokens

- Roughly 4 characters per token
- Input tokens = Prompt
- Output tokens = AI response

Example

`"Hello world"` ≈ 2 tokens

Audio Tokens

Measured in:

- Audio hours processed

Example

1 hour audio = 1 billable audio hour

Token Usage in Your App

Transcription

Whisper Large V3

- 1 minute audio ≈ $0.00185

Whisper Turbo

- 1 minute audio ≈ $0.00067

Turbo exists because humans wanted AI cheaper *and* faster simultaneously.
A deeply exhausting species.

Chatbot Usage

Average Cost Per Conversation

Approximately:

- `$0.0001 - $0.001` (Convert into Rs)

depending on response size.

Gemini Extraction

Average Request

- 500–2000 tokens

Estimated Cost

- Often free under normal educational usage

User Usage Estimates

Light User

Typical Usage

- 1 hour study/day
- 5–10 chat interactions

Estimated Monthly Cost

- **$2 – $7**

Moderate User

Typical Usage

- 2–3 hours study/day
- 20–30 chats

Estimated Monthly Cost

- **$7 – $13**

Heavy User

Typical Usage

- 150+ hours audio/month
- 50+ chats

Estimated Monthly Cost

- **$17+**

Still cheaper than most textbooks. Which is somehow both impressive and depressing.

Database Usage Estimate

Example

A class of:

- 100 students would still remain comfortably inside:
- Supabase free-tier MAU limits

Subscription & Upgrade Options

Groq

- No free subscription plan
- Pure pay-as-you-go

Gemini

- Free tier sufficient for most student projects
- Heavy usage may cost:
    - $30–50/month

Supabase Plans

Plan	Cost
Pro	$25/month
Team	$599/month
Enterprise	Custom Pricing

Recommendations

Suggested Approach

1. Start With Free Tiers

Enough for:

- Development
- MVPs
- Student demos

2. Monitor Usage

Especially:

- Groq API costs

3. Optimize Prompts

Shorter prompts:

- Reduce token usage
- Improve latency

4. Use Whisper Turbo

Approximately:

- 70% cheaper

5. Upgrade Gradually

Scale infrastructure only when needed.

A radical concept in software engineering:
not burning money before users even exist.

Cost Optimization Tips

- Use shorter audio clips
- Batch multiple tasks together
- Cache repeated queries
- Monitor dashboards regularly
- Configure billing alerts

Final Architecture Summary

Layer	Technology
Frontend	React/Vite + Vercel
Backend	FastAPI + Render
Authentication	Supabase Auth
Database	Supabase PostgreSQL
Transcription	Groq Whisper
AI Chat	Groq Llama + Gemini
Data Extraction	Gemini 2.5 Flash

Final Note

Our current stack is actually well-balanced for:

- Student-scale deployment
- MVP demonstrations
- AI-assisted educational workflows

We avoided several classic beginner mistakes:

- Heavy local inference
- Self-hosted GPU nightmares
- Storing files on ephemeral free-tier disks like an optimistic raccoon

So structurally, this is solid. Which is annoyingly rare 😜.

THE END